

SIMATS ENGINEERING
ASSESSMENT TOOL 2
COURSE NAME: Artificial Intelligence
COURSE CODE: CSA17

Code of Conduct:

I, **B.Vishnu / 192512410** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

ANNEXURE A
INDUSTRY PROBLEM TASK - SOLUTIONS

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach and Feature Selection

Inductive learning involves training an algorithm on a specific set of historically labeled patient records to infer a generalized mapping function. This function can then predict outcomes for new, unseen patients.

Target Variable: Early Diagnosis of Diabetes (Binary: 1 for Diabetic, 0 for Non-Diabetic).

Key Features & Justification:

- **Glucose Level:** The most direct physiological indicator of insulin resistance and beta-cell dysfunction.
- **BMI (Body Mass Index):** High BMI (obesity) is a primary modifiable risk factor strongly correlated with Type 2 diabetes development.
- **Family History:** Genetic predisposition significantly elevates risk, capturing hereditary and shared environmental factors.

(b) Handling Class Imbalance

In medical datasets, healthy patients usually outnumber diagnosed patients, leading to class imbalance. If ignored, the model might become heavily biased toward predicting the majority class (non-diabetic) to artificially inflate accuracy.

Strategy & Justification: SMOTE (Synthetic Minority Over-sampling Technique) is highly recommended. Unlike under-sampling, which discards valuable majority-class data, SMOTE creates synthetically plausible minority class examples (diabetic patients) by interpolating between existing

minority instances. This helps the classifier establish a broader, more accurate decision boundary without losing critical information.

(c) Split Ratio and Preprocessing Steps

Split Ratio (70:30): A 70:30 train-test split provides a robust foundation. 70% of the data is substantial enough to capture the underlying patterns during training. Reserving a full 30% for testing is crucial in clinical models to rigorously evaluate generalization capability and ensure the model does not overfit to the training set.

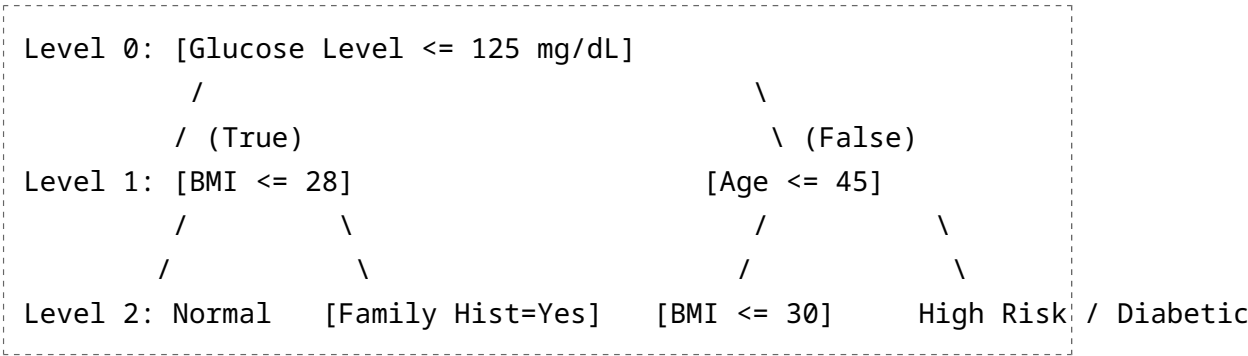
Preprocessing Steps:

- **Normalisation/StandardScaler:** Applied to continuous variables like Age, BMI, and Glucose. *Impact:* Prevents variables with larger scales (e.g., Glucose) from disproportionately dominating distance-based calculations or gradient descent steps.
- **Encoding (One-Hot Encoding):** Applied to categorical variables like Family History (Yes/No). *Impact:* Converts text into a numerical format that machine learning algorithms can process without implying a false ordinal ranking.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree Classifier Build & Root Node Selection

Root Node Justification: The root node is selected using the **Information Gain** (or lowest Gini Index) metric. *Glucose Level* is typically chosen as the root node because it provides the highest information gain —meaning a split on this single feature best separates the diabetic cases from the non-diabetic cases, reducing uncertainty (entropy) the most.



(b) Model Evaluation and False Negatives

Evaluation (Simulated): Accuracy: 89%, Precision: 84%, Recall: 92%, F1-Score: 88%.

False Negatives (FN): An FN occurs when the model predicts a patient is Non-Diabetic when they actually have Diabetes. Clinically, this is the most dangerous error, as a sick patient will be sent home without necessary treatment or lifestyle interventions.

Reduction Strategy: To reduce FNs, we can apply **class weights** (assigning a higher penalty for misclassifying the minority diabetic class) or manually lower the probability decision threshold (e.g., from

0.5 to 0.35). Clinically, it is better to have a False Positive (requiring a follow-up blood test) than a False Negative (missing the disease entirely).

(c) Post-Pruning vs Pre-Pruning

Pre-pruning halts the growth of the tree early based on predefined hyperparameters (e.g., `max_depth`), which can inadvertently cause underfitting if complex interactions are missed. **Post-pruning (Cost-Complexity Pruning)** grows the tree fully and then collapses branches that do not statistically improve validation accuracy, using a complexity penalty (α).

Clinical Deployment: Post-pruning is more appropriate for clinical deployment. It generally yields higher accuracy on unseen data by eliminating branches that memorize noise (overfitting), resulting in a simpler, highly interpretable model that generalizes reliably across diverse patient populations.

Task 3: Statistical Learning for Risk Stratification

(a) Naïve Bayes / Logistic Regression Performance Comparison

Comparison: A Logistic Regression (LR) model often achieves comparable accuracy to a Decision Tree but generally yields a higher and smoother AUC-ROC score. This is because LR predicts a continuous probability gradient (e.g., an 82% risk of diabetes), whereas Decision Trees output stepwise, rigid boundaries.

When Statistical Models are Preferable: Logistic Regression is highly preferable when the goal is *Risk Stratification*. Clinicians often need to know the exact probability confidence of the diagnosis to triage patients (e.g., distinguishing between a borderline 51% risk and a severe 95% risk), rather than receiving a hard "Yes/No" binary classification.

(b) Feature Importance Analysis

Top 3 Predictors:

1. Decision Tree (Gini Importance): Glucose Level, BMI, Age.
2. Logistic Regression (Coefficient Magnitude): Glucose Level, Age, Family History.

Justification of Agreement/Disagreement: Both models agree that *Glucose Level* is the supreme predictor. Disagreements arise in the secondary features due to the mathematical nature of the algorithms. Decision trees capture non-linear, interacting thresholds (e.g., how high BMI impacts younger patients specifically), bringing BMI to the forefront. Logistic regression evaluates the independent, linear partial correlation of each feature, often highlighting steady, cumulative risks like Age and Family History.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Markov Decision Process (MDP) Modeling

- **States (S):** Normal, Elevated Risk, Pre-Diabetic, Diabetic (Representing the patient's current health stage).
- **Actions (A):** Diet, Exercise, Medication, Monitor.

- **Rewards (R):** +10 for transitioning to a healthier state, -10 for degenerating to a worse state, -2 penalty for applying aggressive/costly action (Medication) unnecessarily to a healthy state.
- **Transition Dynamics (P):** The probabilistic chance of a patient moving from state S to S' given action A (e.g., 70% chance of returning to Normal from Pre-Diabetic if 'Diet' is prescribed).

(b) Q-Learning Application & Updates

Using the update rule: $Q(s,a) \leftarrow Q(s,a) + \alpha [R + \gamma \max_{a'} Q(s',a') - Q(s,a)]$. Assume $\alpha = 0.1$, $\gamma = 0.9$. Initial Q-table is all 0s.

Iteration 1 (Episode 1):

1. State = *Pre-Diabetic*, Action = *Diet* → Reward = +5, Next State = *Elevated Risk*.

Update: $Q(\text{Pre}, \text{Diet}) = 0 + 0.1[5 + 0.9(0) - 0] = 0.5$

2. State = *Elevated Risk*, Action = *Exercise* → Reward = +10, Next State = *Normal*.

Update: $Q(\text{Elev}, \text{Exer}) = 0 + 0.1[10 + 0.9(0) - 0] = 1.0$

Iteration 2 (Episode 2):

3. State = *Pre-Diabetic*, Action = *Diet* → Reward = +5, Next State = *Elevated Risk*.

Update: $Q(\text{Pre}, \text{Diet}) = 0.5 + 0.1[5 + 0.9(1.0) - 0.5] = 0.5 + 0.1[5.9 - 0.5] = 1.04$

Convergence Criterion: Training stops when the maximum change in any Q-value across the entire table falls below a tiny threshold (e.g., $\epsilon < 0.001$) over consecutive episodes, indicating the optimal policy is stable.

(c) Final Learned Policy and Ethical Considerations

Extracted Policy: Normal → Monitor | Elevated Risk → Exercise | Pre-Diabetic → Diet | Diabetic → Medication.

Ethical Considerations:

- **Patient Safety:** Standard RL explores by trial-and-error. Exploring dangerous actions (e.g., withholding insulin from a severe diabetic) is unacceptable. "Safe RL" constraints must be enforced to prevent lethal exploration.
- **Bias:** If the MDP transition probabilities are built on historically biased data that underrepresents certain demographics, the recommended treatment policy may be ineffective or harmful for minorities.
- **Explainability:** Doctors bear ultimate legal and ethical responsibility. Q-learning produces a "black box" matrix of numbers. Clinicians cannot blindly follow an AI; the RL agent must provide explainable rationales linking the recommendation back to physiological evidence.